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Project Title:	NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL		
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Comments:			
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
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Task Description:	NOTE: Continuation of "NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL," grant number NNX10AU19G. Funding is for Dr. Kelton's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main project: Thermophysical and structural properties and solidification behavior of supercooled glass forming liquids as a member of THERMOLAB; the Science Coordinator is Prof. Dr. Hans Fecht, Ulm University, Institute of Micro and Nanomaterials, Albert-Einstein-Allee 47, Ulm Germany. Similar measurements on undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); the Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum fur Luft- und Raumfahrt (DLR), Institut fur Materialphysik im Weltraum, 51170 Köln, Germany. Subprojects: 1. Ground-Based Experimental Program: Synchrotron X-ray studies of liquid structures are made at the Advanced Photon Source using the ESL (electrostatic levitation) chamber at Washington University (WU-BESL), which has been optimized for scattering studies. These are correlated with neutron scattering data taken at the Spallation Neutron source using the Neutron ESL (NESL), with thermophysical property data obtained on ground with WU-BESL and using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) on the International Space Station (ISS). Terrestrial maximum undercooling and induction time to recalescence data are obtained in ESL, where stirring ensures that nucleation is not diffusion controlled. This data will allow an evaluation of diffusion in the upcoming MSL-EML ISS undercooling studies. 2. Thermophysical Property Measurements: ISS. Specific heat, density, surface tension, viscosity, and electrical resistivity measurements are made on supercooled glass-forming liquids and of Ti-Zr-Ni liquids with different Ni concentrations as a function of the temperature using the MSL-EML facility. These are correlated with x-ray synchrotron and elastic neutron diffraction studies of the supercooled liquids and ground-based measurements made using WU-BESL The purpose is to determine the existence and nature of a cooperative cross-over behavior in the shear viscosity at high temperature, which may be associated with chemical and topological ordering and a possible connection with liquid fragility. Measurements of the specific heat and electrical resistivity are also of interest for their correlation with the dynamical crossover. 3. Effect of diffusion on nucleation: The importance of long-range diffusion on nucleation is being evaluated using the MSL-EML facility located on the ISS. The critical items of interest are the maximum undercooling and the induction time to recalescence in isothermal holds at temperatures below the liquidus temperature. Together with the results from the thermophysical property measurements and an MHD convection model results this will allow us to analyze how the maximum undercooling is influenced by fluid flow. These results of the micro-g experiments are compared with the results of ground-based ESL measurements and with predictions from Kelton’s coupled-flux model for nucleation. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: 1. Ground Based Experiment Program. Measurements of the density, surface tension and viscosity, and synchrotron x-ray diffraction measurements of the liquid structure in the supercooled state using WU-BESL. Ground-based measurements of the maximum undercooling and induction time to recalescence in isothermal holds are also made using WU-BESL. These will be used to evaluate the nature of ordering on phase transitions in the supercooled liquid and fragility. 2. Long Duration Microgravity Experiments. Measurements of thermophysical properties, specific heat (modulated measurements), surface tension, viscosity and density, and also electrical resistivity are made as a function of temperature using the MSL-EML on the ISS. The results are coupled with x-ray and elastic neutron diffraction measurements of the liquid structure factor (WP 2) and ground based studies of the density and viscosity to study the role of ordering on novel liquid/liquid phase transitions and to determine the nature of these transitions. Maximum undercooling and time to recalesce measurements as a function of temperature and stirring will be compared with terrestrial ESL measurements to assess the importance of diffusion on nucleation, allowing Kelton's coupled flux model for nucleation to be evaluated experimentally.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which have the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science, encapsulated in the Materials Genome Initiative. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. Even containerless based methods, such as electrostatic levitation, cannot be used to accurately investigate these. The ISS experiments that are being carried out under the support of this grant are allowing these data to be obtained.

Task Progress:

Summary of activities during 2018-2019
1. Background

This research is focused on two key questions. The first is the investigation of how to describe crystal nucleation from a liquid if the two phases have a different chemical composition. The second explores how the high temperature thermophysical properties of the liquid may inform glass formation. To address these questions, experiments are being made on samples with no containers using the microgravity environment of the International Space Station and the technique of electrostatic levitation under terrestrial conditions. A summary of the results obtained for 2018 – 2019 is provided in this report.

Nucleation is the first step in the crystallization of a liquid. An improved understanding of this process is critical for optimizing materials to particular applications and for glass formation. It is most often described using the classical theory of nucleation (CNT), developed in the late 1800s and early 1900s to explain gas condensation and extended to describe crystal nucleation in a liquid around 1950. The CNT assumes that the kinetics of nucleation are controlled by how quickly an atom in the liquid attaches itself to the crystal nucleus. Viewed from the perspective of how the number of atoms increases in the cluster with time, this stochastic interfacial attachment process is analogous to a diffusion process. For the nucleation of a crystal phase from a liquid with a different chemical composition, the kinetic picture can become much more complex. Then the interfacial attachment couples to diffusion within the liquid, leading to a coupling of two stochastic processes. The coupled flux model was developed by the Principal Investigator (PI) Kelton to deal with this. Testing this model is a key goal of the ISS and terrestrial investigations.

Turning to the second question, experiments have shown that the temperature dependence of the liquid viscosity abruptly changes at a temperature TA, located near the melting temperature. Molecular dynamics (MD) studies indicate that this is due to the onset of cooperative structural rearrangements within the liquid. The structural changes should be revealed in the specific heat and electrical resistivity of the liquid. The ISS offers a unique facility for measuring these quantities, which can then be related to the measured viscosity as a check of the MD predictions. Our work also suggests that the processes that begin at TA culminate in glass formation at the glass transition temperature, Tg. If so, it should be possible to use the high temperature properties of the liquid to estimate Tg and even to predict glass formation.

2. ISS experiments

Nucleation Studies – As stirring in the liquid decreases, atomic transport in the liquid becomes more dominated by long-range diffusion. For this case, the coupled flux model predicts a decrease in the nucleation rate. The nucleation rate can be deduced from the maximum amount that the liquid can be cooled below the melting temperature (maximum undercooling) before crystal formation. To investigate the predications of the coupled flux model, the maximum undercooling was measured in Vit 106 (a good metallic glass forming liquid) and in a Cu50Zr50 liquid, both as a function of stirring. The changes in stirring were made by adjusting the voltages on the heater and positioner coils in the electromagnetic levitation facility (MSL-EML) on the ISS. In both cases, an increase in stirring gave a smaller value for the maximum undercooling, in agreement with predictions. The coupled flux model predicts that the nucleation pre-factor, A*, should become larger as the transport due to stirring becomes larger than that due to diffusion. The nucleation values for A* obtained from the maximum undercooling data for both liquids showed this to be true. These results demonstrate for the first time that fluid flow affects nucleation as has been predicted by the coupled-flux model for nucleation.

Electrical Resistivity – Measurements of the thermophysical properties of several liquids using the MSL-EML were reported in last year's report. Of particular note were the first measurements of the electrical resistivity as a function of temperature for these liquids. A detailed analysis of those resistivity data over the past year has led to exciting new results. Within experimental error, upon cooling, the change in the resistivity begins to increase near the dynamical crossover temperature, TA, which was discussed in the background section of this report. Based on the Faber-Ziman theory, the most accepted theory for the electrical resistivity of a liquid, the resistivity is a function of the structure factor, S(q). Our ground-based synchrotron X-ray scattering studies have demonstrated that S(q) sharpens monotonically with decreasing temperature, indicating structural ordering in the liquid. The sudden change in the electrical resistivity at TA, then, establishes a direct connection between structure and dynamics, as predicted from the MD studies.

The data also show that the electrical resistivity saturates at temperatures above TA. Although saturation of the resistivity in highly disordered solids (especially in glasses at low temperatures and crystalline solids at high temperatures) is known, this is the first experimental demonstration of such a phenomenon in a metallic liquid. Since S(q) is temperature-dependent below and above TA, the saturation cannot reflect a saturation in the ordering of the liquid. Instead, it likely points to ineffective electron-phonon scattering above this temperature, which is not explained within existing theories of the electrical resistivity in metallic liquids.

3. Other ISS-related activities

The experience gathered from the past experiments on the ISS has led to the development of new strategies for more effective experiments for the future runs on the ISS using the MSL-EML facility. Activities over the past year aimed at those future studies include the following.

(a) An additional measurement campaign (2.4) in early 2020 to complete experiments on a batch 2 sample (Ti39.5Zr39.5Ni21) by obtaining statistically significant data on the influence of stirring on nucleation. Forty undercooling cycles will be made under vacuum, with half of them at minimum stirring (heater off, minimum positioner voltage) and the other half at maximum possible stirring (heater 0.2 V and minimum positioner voltage).

(b) Batch 3 samples – QUASI is the PI for two alloys, Cu47Zr47Al6 and Zr80Pt20. It is co-PI with ICOPROSOL for Ti45Zr45Ni10 and with THERMOLAB for Pt57.5Cu14.7Ni5.3P22.5. All of these alloys were selected for study in in batch 3. The Cu47Zr47Al6, Zr80Pt20 and Ti45Zr45Ni10 flight samples were prepared by QUASI and delivered to AIRBUS in September 2019. The Pt57.5Cu14.7Ni5.3P22.5 sample is provided by THERMOLAB. QUASI will measure the maximum undercooling with stirring and the electrical resistivity of these liquids.

(c) Batch 4 – Proposals were submitted for batch 4 in July 2019. A Zr50Pd50 alloy was proposed by QUASI in partnership with MULTIPHAS, THERMOPROP, ICOPROSOL, and USTIP. QUASI and ICOPROSOL also proposed a Ti40Zr27Ni33 Laves phase-forming alloy to continue their efforts to understand the effect of stirring on nucleation. A quasicrystal-forming liquid (Ti39.5Zr39.5Ni21) was studied in batch 2 and a cubic solid solution forming liquid (Ti45Zr45Ni10) is scheduled to be studied in batch 3. In a continuation of a joint effort to understand nucleation and

growth of crystals from supercooled liquids QUASI and MULTIPHAS proposed a bulk metallic glass forming Zr50Cu30Ni20 liquid alloy for batch 4.

4. Ground-based activities

Ground-based structural and maximum undercooling studies of liquids give complimentary information to the ISS experiments. This past year we have focused on obtaining statistically significant nucleation data by making hundreds of maximum undercooling measurements on each liquid composition and analyzing the data using a method developed by Skripov. The derived interfacial free energy, s, which gives a measure of the nucleation barrier, varied among the liquids studied. The smallest interfacial energy for the Ti39.5Zr39.5Ni21 is due to very similar short-range-order (SRO) in the liquid and the quasicrystal that nucleates first from the liquid, as was demonstrated from our earlier structural studies. The interfacial free energy increases for the Laves phase-forming liquid, Ti40Zr30Ni30, since the local order of the nucleating solid has some similarity with the order in the liquid, but differs in detail. Finally is was largest for Ti45Zr45Ni10 since the local order of the nucleating cubic ß-solid solution phase is very different from that in the liquid [Lee et al., 2005]. ESL-based studies provide the most quiescent conditions for nucleation studies, when the laser is turned off stirring quickly decays. Additional measurements are, therefore, being made on other batch 2, batch 3, and proposed batch 4 samples to provide complimentary data to that obtained using the MSL-EML on the ISS.

A second significant activity this past year was the continued investigation of whether high-temperature liquid properties can predict glass formation. Over the course of this grant and a complementary National Science Foundation (NSF) grant (DMR 15-06553), we have identified a number of important correlations between the glass-transition temperature, T_g, dynamical crossover temperature, T_A, thermal expansion coefficient, alpha, liquid fragility, m , rate of change of the first peak height/width of the structure factor with temperature, (dS(q,T))/dT, and glass formability of metallic liquids. We have demonstrated that by knowing the viscosity and the thermal expansion coefficient of the liquid near the equilibrium melting temperature, the value of Tg can be accurately predicted, agreeing to within 2% of the measured value. Glass formation can also be predicted more accurately than within other approaches. One publication has appeared in print and another is in press, which describe these results. The focus on most of the glasses thus far have been on the thermophysical and dynamic properties near Tg; our results indicate that the properties above the liquidus temperature are equally, or perhaps more, important. Most importantly, they suggest that the glass transition actually starts at TA. These results could possibly lead to a paradigm shift in the prevalent ideas on glass formation.

Reference

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Bibliography Type:	Description: (Last Updated: 05/20/2025)
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